Quiz 6

Conditional distributions, covariance, and averages

Try each question before reading the worked solutions on page 2.

1. $f_{X,Y}(x,y) = 2$ on $0 < x < y < 1$. Given $Y = 0.8$, what is $P(X < 0.2 \mid Y = 0.8)$?

A. 0.2

B. 0.25

C. 0.4

D. 0.8

2. $f_{X,Y}(x,y) = 2$ on $0 < x < y < 1$, zero elsewhere. What is $\mathbb{E}[X \mid Y = y]$ for $0 < y < 1$?

A. $1/2$

B. y

C. $y^2/2$

D. $y/2$

3. $\mathrm{Var}(X) = 2$, $\mathrm{Var}(Y) = 3$, $\mathrm{Cov}(X,Y) = 1$. What is $\mathrm{Var}(X - Y)$?

A. 3

B. 5

C. 7

D. 1

4. Independent, identically distributed observations have mean 5 and standard deviation 4. For a sample of size $n = 64$, what is the standard deviation of their average?

A. 4

B. $1/16$

C. $1/2$

D. 32

5. For $N \sim \mathrm{Bin}(100, 0.2)$, what are its mean and variance, respectively?

A. 20 and 20

B. 20 and 16

C. 80 and 16

D. 20 and 4

Worked solutions

1. A conditional probability from a density

Answer B

Fixing $Y = 0.8$ leaves $0 < x < 0.8$. First find the marginal density:

$$f_Y(0.8) = \int_0^{0.8} 2\,du = 1.6.$$

Then $f_{X|Y}(x \mid 0.8) = 2/1.6 = 1/0.8$ for $0 < x < 0.8$, and zero elsewhere. Integrate this conditional density over $0 < x < 0.2$:

$$P(X < 0.2 \mid Y = 0.8) = \int_0^{0.2} \frac{1}{0.8}\,du = \frac{0.2}{0.8} = 0.25.$$

The denominator is the density $f_Y(0.8)$, not $P(Y = 0.8)$.

2. Conditional mean

Answer D

For $0 < y < 1$, $f_Y(y) = \int_0^y 2\,du = 2y > 0$. Hence $f_{X|Y}(x \mid y) = 1/y$ on $0 < x < y$, and zero elsewhere. Weight each value by this conditional density:

$$\mathbb{E}[X \mid Y = y] = \int_0^y u\,\frac{1}{y}\,du = \frac{1}{y}\left[\frac{u^2}{2}\right]_0^y = \frac{y}{2}.$$

This is the midpoint of the conditional uniform interval $(0, y)$.

3. Variance of a difference

Answer A

For a linear combination, the covariance term is $2ab\,\mathrm{Cov}(X, Y)$. With $a = 1$ and $b = -1$,

$$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y) = 2 + 3 - 2(1) = 3.$$

The minus sign comes from the product of the two coefficients. The covariance cannot be omitted here.

4. Standard deviation of an average

Answer C

Write $\overline{X} = n^{-1}\sum_{i=1}^n X_i$. Independence makes the covariances zero. Since each variance is $4^2 = 16$,

$$\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(X_i) = \frac{n(16)}{n^2} = \frac{16}{64} = \frac{1}{4},$$

$$\mathrm{SD}(\overline{X}) = \sqrt{\mathrm{Var}(\overline{X})} = \sqrt{\frac{1}{4}} = \frac{1}{2}.$$

The question asks for standard deviation, so take the square root of the variance.

5. Binomial mean and variance

Answer B

A binomial count is a sum of n independent Bernoulli indicators, each with mean p and variance $p(1-p)$. Their means and variances add:

$$\mathbb{E}[N] = np = 100(0.2) = 20, \qquad \mathrm{Var}(N) = np(1-p) = 100(0.2)(0.8) = 16.$$

The value $4 = \sqrt{16}$ is its standard deviation, not its variance.